

Question B2

This question carries 18 marks in total.

(a) [3 MARKS] Find a scalar potential for the following vector field:

$$\mathbf{F}(x, y, z) = (2x + ze^{xz}, z, y + xe^{xz})$$

(b) [3 MARKS] Let $f(x, y, z) = x + ze^{-xy} - 1 = 0$ define a surface. Find the general form for the slope of the surface in the direction of the vector $\mathbf{v} = (0, -1, 1)$.

(c) [6 MARKS] Find the angle between the normals to the surfaces

$$\sin(x) + \cos(y) - \frac{xy}{\pi} + z = 2 \quad \text{and} \quad z = x^2 - (y - \pi)^2 + 1$$

at the point $(\pi, 0, 1)$.

(d) [3 MARKS] Given $\mathbf{F} = yz\mathbf{i} + 5xz\mathbf{j} - (x - y - z)\mathbf{k}$ find $\nabla \times \mathbf{F}$.

(e) [3 MARKS] Calculate the volume of the solid bounded by $z = 2 - x^2 - y^2$ and $z = \sqrt{x^2 + y^2}$.